

# Quotient Laws of Exponents

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## MA.7.NSO.1.1

*I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.*

*Yesterday: same base, add the exponents.*

**Today: same base, subtract — and exponent 0 is not 0.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1. Write as one power:  $3^2 \cdot 3^3$
2.  $\sqrt[3]{-27}$
3. Solve  $2x + 3 < 11$
4. Simplify  $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

# Warm-Up

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*Answers.*

1. Write as one power:  $3^2 \cdot 3^3$   $3^5$

2.  $\sqrt[3]{-27}$   $-3$

3. Solve  $2x + 3 < 11$   $x < 4$

4. Simplify  $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$   $5$

# Notes

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*Where the rule comes from.*

## I. Quotient of powers — same base, subtract the exponents

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$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

**Quotient of powers:**  $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values:  $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

# Notes

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## II. Power of a quotient — the exponent goes top and bottom

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$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

**Power of a quotient:**  $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way:  $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$  and  $(0.8)^3 = 0.512$

# Notes

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## III. The zero exponent — two roads to the same answer

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| $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|-------|-------|-------|-------|-------|
| 16    | 8     | 4     | 2     | 1     |

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

**Zero exponent:**  $a^0 = 1$  for every nonzero  $a$

# Example 1

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*A negative base.*

$$\frac{(-2)^6}{(-2)^3}$$

**Which law? Then find the value.**

# Example 1

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*Worked.*

$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3 \quad \text{quotient of powers: subtract}$$

$$(-2)^3 = -8 \quad \text{three negative factors}$$

$$64 \div (-8) = -8 \quad \text{check: evaluate each, then divide}$$

**Answer: -8**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\frac{3^6}{3^2}$$

# Your Turn

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*Answer.*

$$\frac{3^6}{3^2}$$

same base, subtract:  $3^4$

**Answer:  $3^4 = 81$**

# Example 2

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*A decimal base inside a quotient.*

$$\left(\frac{0.2}{10}\right)^4$$

**Find the value.**

# Example 2

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*Worked both ways.*

$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000} \quad \text{power of a quotient}$$

$$\frac{0.0016}{10000} = 0.00000016 \quad \text{divide by } 10^4: \text{ four places left}$$

$$(0.02)^4 = 0.00000016 \quad \text{check: divide first, then raise}$$

**Answer: 0.00000016**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\left(\frac{3}{4}\right)^3$$

# Your Turn

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*Answer.*

$$\left(\frac{3}{4}\right)^3$$

numerator and denominator both cubed

$$\textbf{Answer: } \frac{27}{64}$$

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$\frac{10^5}{10^3}$$

**Answer it.**

*Write it as one power.*

# Whiteboards • Question 1 of 9

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*Answer.*

$$\frac{10^5}{10^3}$$

same base, subtract the exponents

**Answer:  $10^2$**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

A website puts  $8^4$  photos into 8 equal albums.

The site is 30 days old.

**How many photos are in each album?**

**Answer it.**

# Whiteboards • Question 2 of 9

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*Answer.*

A website puts  $8^4$  photos into 8 equal albums.

The site is 30 days old.

**How many photos are in each album?**

$$8^4 \div 8^1 = 8^3$$

**Answer: 512**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$\frac{(-6)^6}{(-6)^5}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 3 of 9

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*Answer.*

$$\frac{(-6)^6}{(-6)^5}$$
$$(-6)^1$$

**Answer: -6**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$\left(\frac{-5}{10}\right)^7$$

**Answer it.**

*Apply the power of a quotient law. Do not simplify.*

# Whiteboards • Question 4 of 9

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*Answer.*

$$\left(\frac{-5}{10}\right)^7$$

the exponent goes to both

$$\textbf{Answer: } \frac{(-5)^7}{10^7}$$

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$9^0$$

Which is the value?

**A.** 1

**B.** 9

**C.**  $\frac{1}{9}$

**D.** 0

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

$$9^0$$

Which is the value?

**A. 1**

**B. 9**

**C.  $1/9$**

**D. 0**

**Answer: A — 1**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

**A.**  $1^6$

**B.**  $4^{10}$

**C.**  $4^6$

**D.**  $4^4$

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

**A.**  $1^6$

**B.**  $4^{10}$

**C.**  $4^6$

**D.**  $4^4$

**Answer: C —  $4^6$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

$$\frac{0.5^8}{0.5^5}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 7 of 9

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*Answer.*

$$\frac{0.5^8}{0.5^5}$$
$$0.5^3$$

**Answer: 0.125**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

$$\left(\frac{10}{11}\right)^0$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 8 of 9

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*Answer.*

$$\left(\frac{10}{11}\right)^0$$

any nonzero base to the zero power

**Answer: 1**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Hannah cancels:  $\frac{2^3}{2^3} = \frac{8}{8} = 1$ . Sebastian subtracts:  $\frac{2^3}{2^3} = 2^0$ .

Who is right?

**Write your answer in sentences.**

*Say what each one did, and what it tells you about  $2^0$ .*

# Whiteboards • Question 9 of 9

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*Answer.*

Hannah cancels:  $\frac{2^3}{2^3} = \frac{8}{8} = 1$ . Sebastian subtracts:  $\frac{2^3}{2^3} = 2^0$ .

Who is right?

both are right, so  $2^0$  must equal 1

**Answer: Both —  $2^0 = 1$**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Powers with negative bases — WFW

Division with exponents — XNC

Identify equivalent expressions involving whole-number exponents — KBE